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NPTEL ONLINE CERTIFICATION COURSE

SUSTAINABLE TRANSPORTATION SYSTEMS

Lecture 26: TOD Case Study-I: Section of Delhi Metro

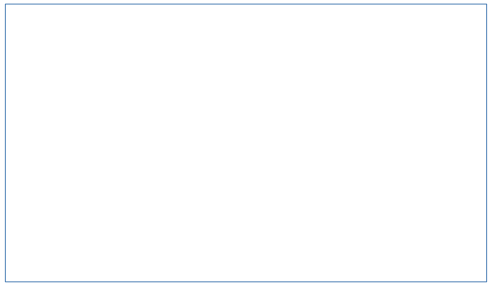
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Contents

TOD planning for Chhatarpur to Arjangarh transit under Delhi metro project.

- Study area attributes
- Existing Land use
- Planned transit corridor & development plan
- Planned land use in area along the transit
- Impact of TOD on various aspects of development
- Cash flow estimation
- Potential risk



Study Zone: Delhi



Delhi: Capital city of India; Union Territory state

- Area- 1483 Sq. Kms
- Population (2011)- 16.7 million
- Vehicles registered (2011)- 6.93 million
- Vehicle- 420 vehicles per 1000 population.
- Modal share (2008)- 45:55 (public: private)
- Modal shift (Target)- 80:20

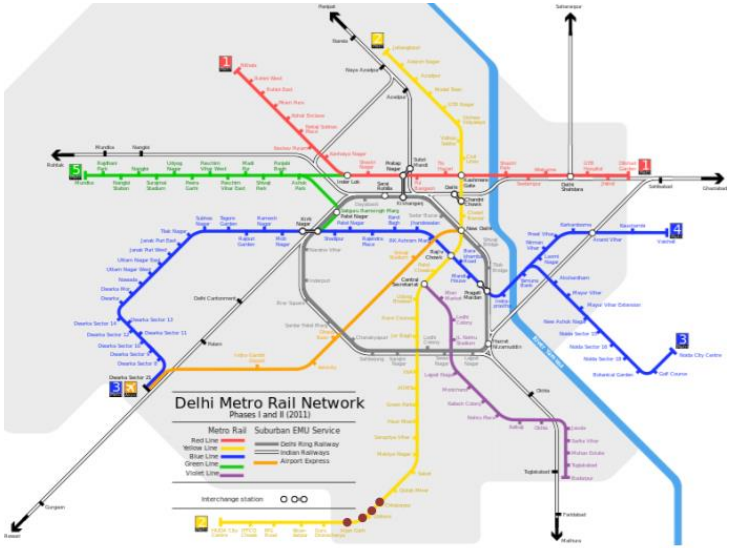
Car dependent society with dispersed activity zones and long-distance trips.

Image source: mapsofindia.com

Delhi Metro

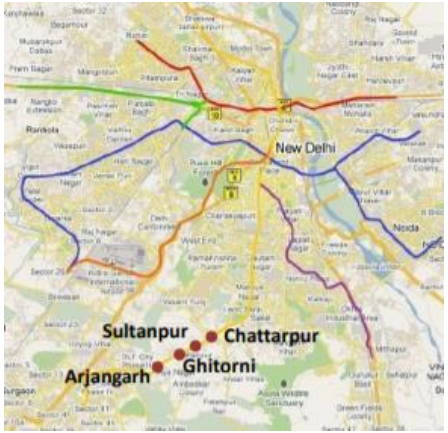
Phase 1 and Phase 2 of Delhi Metro constitutes;

- 6 Lines with 193 kms
- 145 metro stations.
- Connects Delhi with
  - Gurgaon
  - Noida
  - Ghaziabad
  - Faridabad
- Carries 1.6 Million Passengers per day (phase 2 completion)



Study area - metro corridor from Chattarpur to Arjangarh

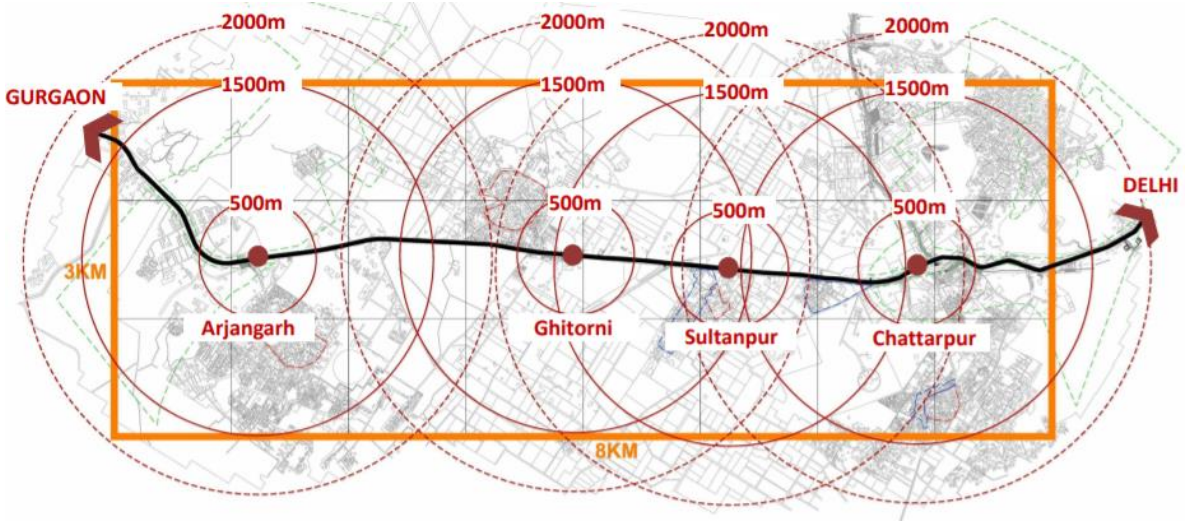
- The Site lies along the Mehrauli-Gurgaon road and is flanked by low density development on either side
- Connector between Gurgaon and Delhi
- This stretch is an entry into Gurgaon through road and as well as the metro



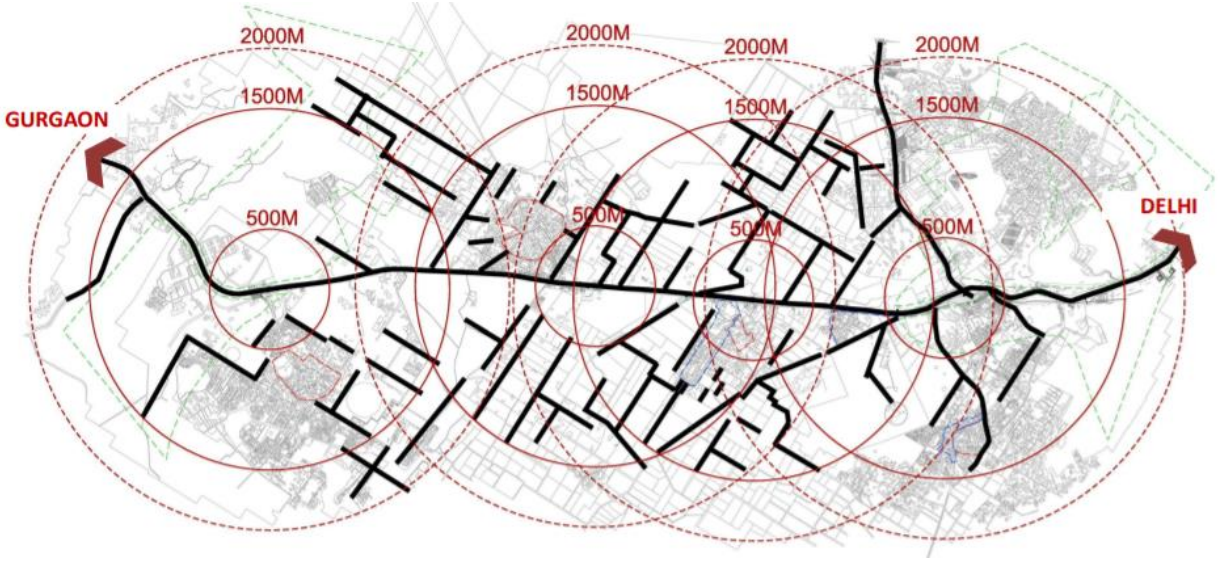


Influence zone of transit

- Total Study Area- 24 sq. Km (2,400 Hectares)
- Walking Influence Zone- 500m (6min)
- Non-Motorized Transport Influence Zone- 1500m (6min)



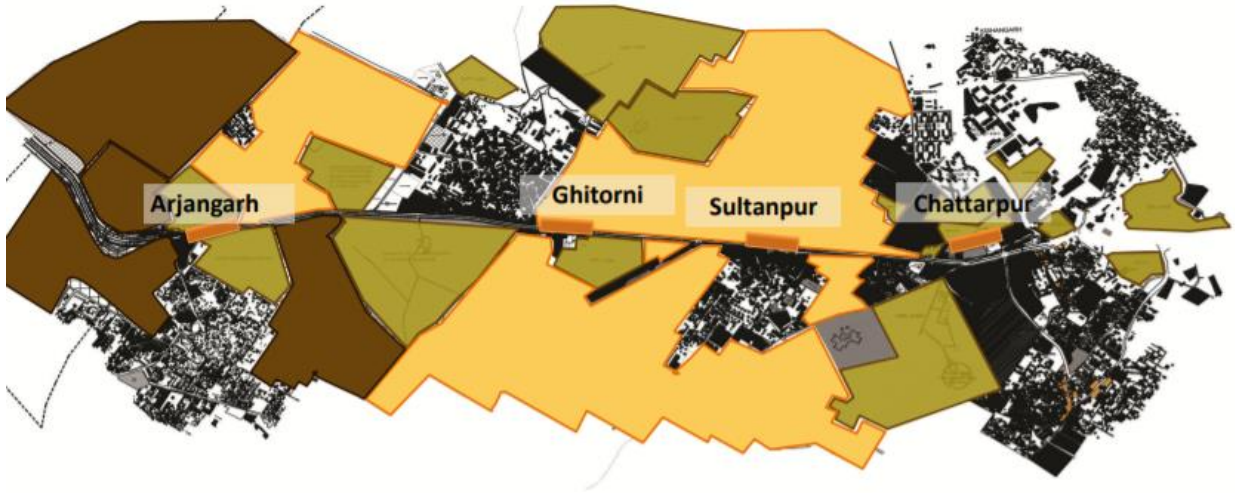
Site attributes - Urban grain



Urban grain: Basic layout of pattern of plots/blocks in any area.

Study area consists incomplete roads, low-density, low-rise development

Site attributes - Hard areas and soft areas

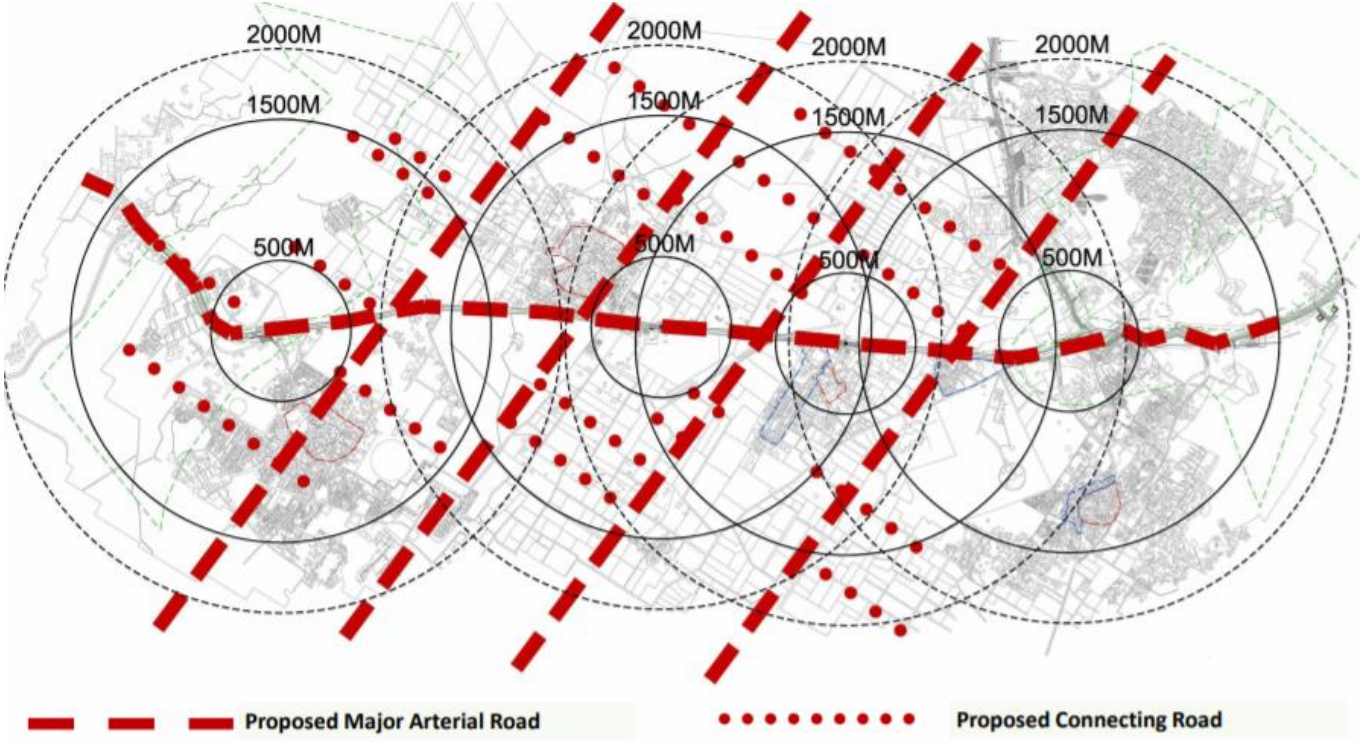


- Hard land: Land use can't be changed
- Soft land: Free land/greenfield development possible
- Moderate: Changes in land use possible

Site attributes: Urban character

- Chattarpur – Cultural and Tourism (Hotels, Marriage gardens etc.)
- Sultanpur – Commercial Village (unplanned area with many commercial activities)
- Ghitorni – Retail (Independent housing with small shops in locality)
- Arjangarh – Many institutions

Transport corridors



- Major arterial road is along the transit and perpendicular to transit at regular interval.
- Connecting roads are also planned for better end-mile connectivity.

Land use development plan



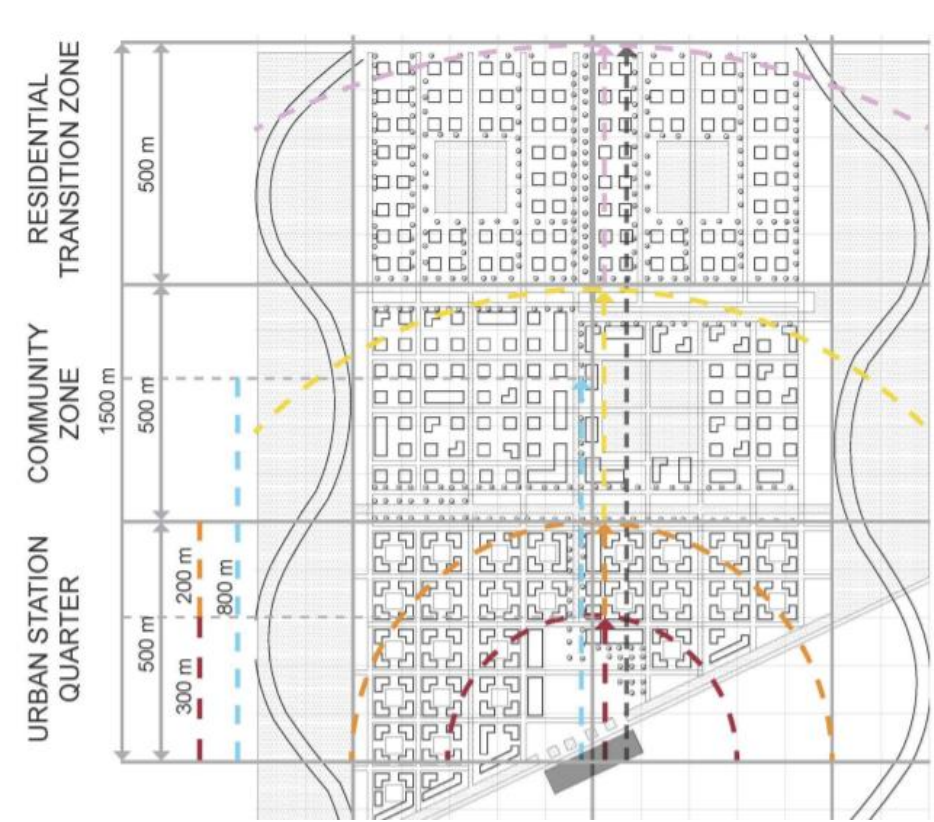
- Land use planning is done for area around the transit.
- Green belts, public parks and open spaces are provided.
- Building blocks are also planned accordingly.

Concept masterplan for study area



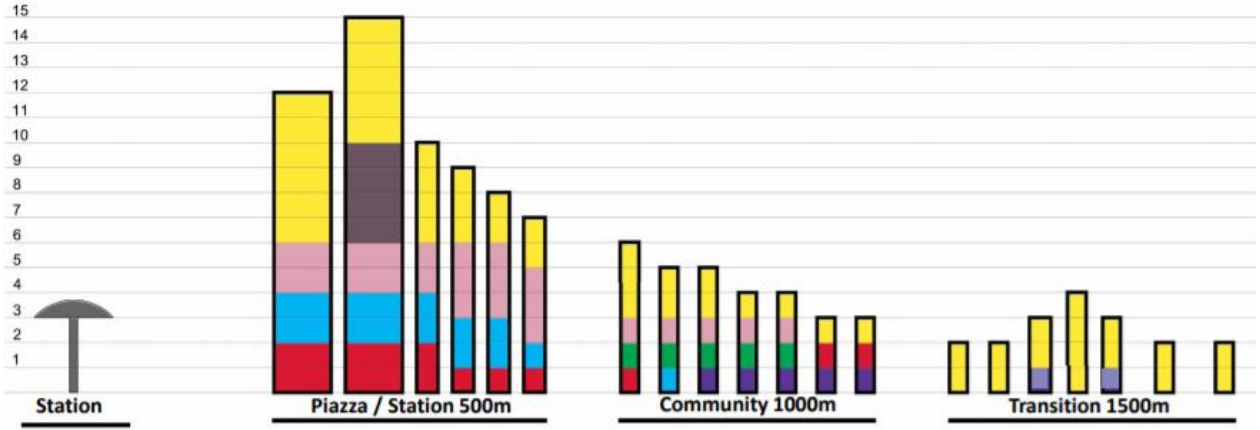
Overall area development plan in study area is shown in the image.

Land use development: Typology



- Typical Station
- 300 m ZONE - High Density
- 200 m ZONE - High Density
- 500 m ZONE - Medium Density
- 500 m ZONE - Low Density
- 500 m ZONE – 6 minute Walk
- 1500 m ZONE – NMT 6 minute
- 800 m ZONE - 10 minute Walk
- 2000 m ZONE - NMT 10 minute

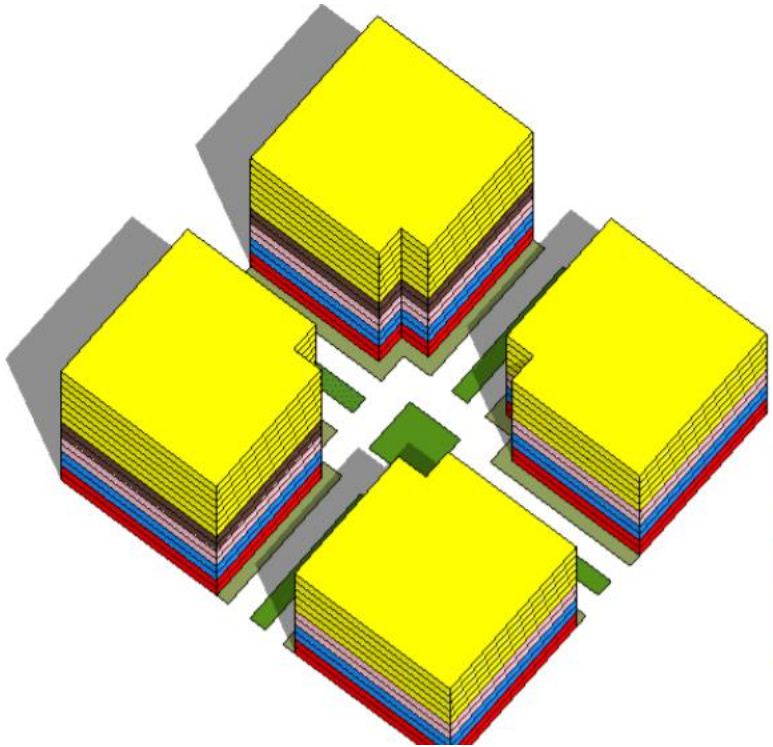
Mix use of land



Overall Development Mix	
Residential	40% - 50%
Hotel	3%- 6%
Commercial	10% - 20%
Entertainment	10% - 12%
Retail	10% - 20%
Community / Social	5% - 6%
Education	5% - 6%
Religion / Cultural	2.5%- 5%

Different uses are planned for station area (with 500 m. of station) and similarly for community and transition zone.

Land-use planning: Station area



6 FAR

500 Du/Ha

Mixed Use

Residential	40%
Hotel	6%
Commercial	10%
Entertainment	10%
Retail	17%
Community / Social	6%
Education	6%
Religion / Cultural	5%

- FAR: Floor area ratio
- Du: Development unit
- Station area is 500 radius area with metro station.

FAR (Floor area ration)

Land/plot coverage: Area % of plot on which construction is permitted.

FAR Example:

If the size of plot is 100 sq. ft. and land coverage is 50% and allowed FAR is 6, then overall construction area will be 600 sq. ft. and for that no. of floors will be 12.

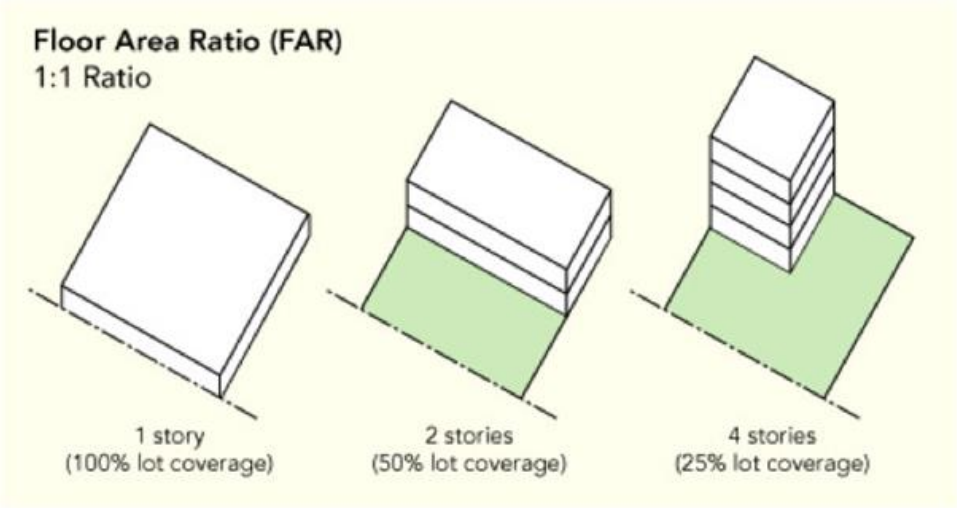
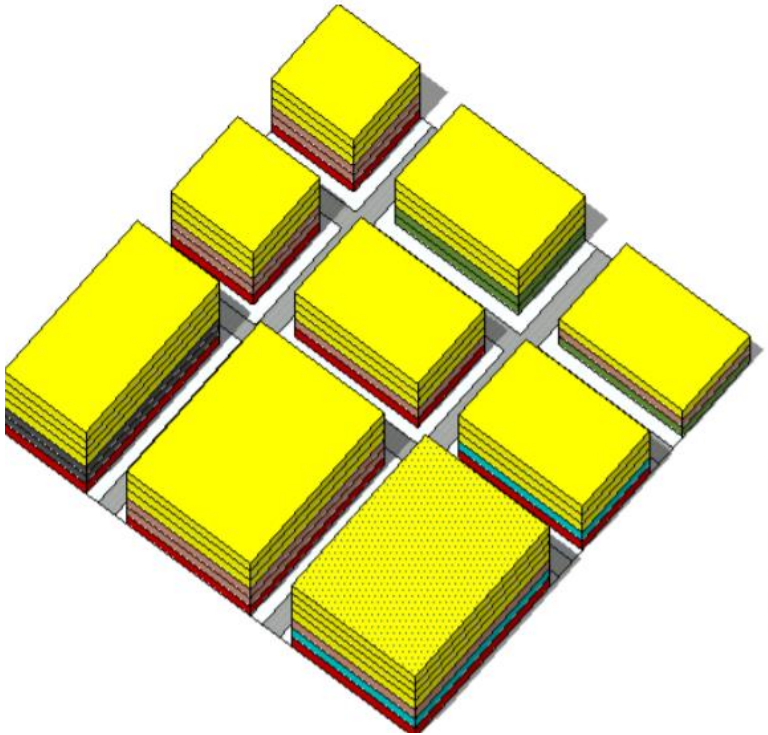


Image source: minnpost.com

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Land-use planning: Community zone



3.5 FAR

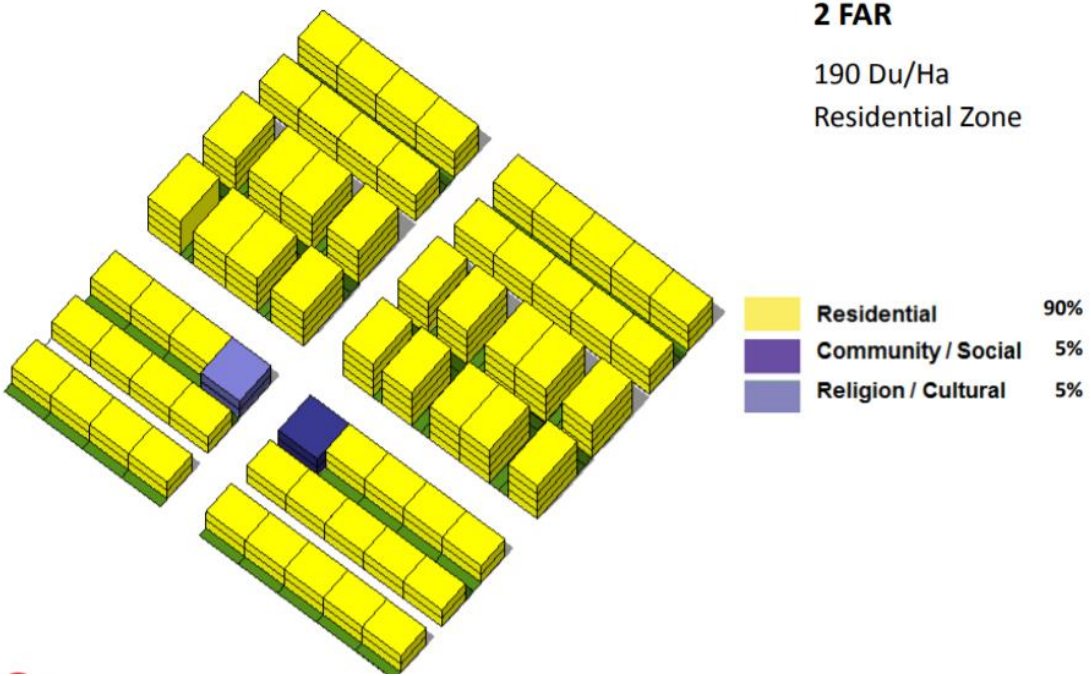
340 Du/Ha

Residential Dominated

Residential	55%
Hotel	3%
Commercial	10%
Entertainment	10%
Retail	8%
Community / Social	6%
Education	6%
Religion / Cultural	2%

Community zone is area between 500m. & 1000 m. radius circle.

Land-use planning: Transition Zone



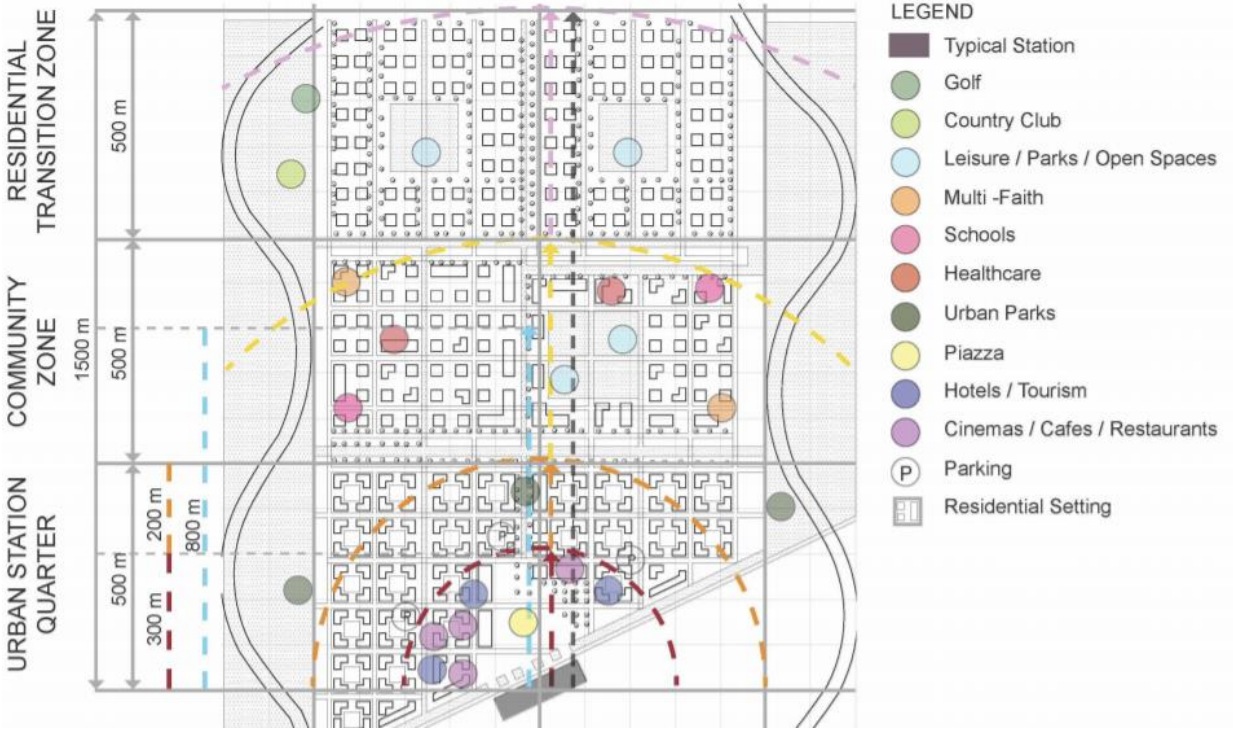
Transition zone is area between 1000m. & 1500 m. radius circle.

Overall model for land use along transit



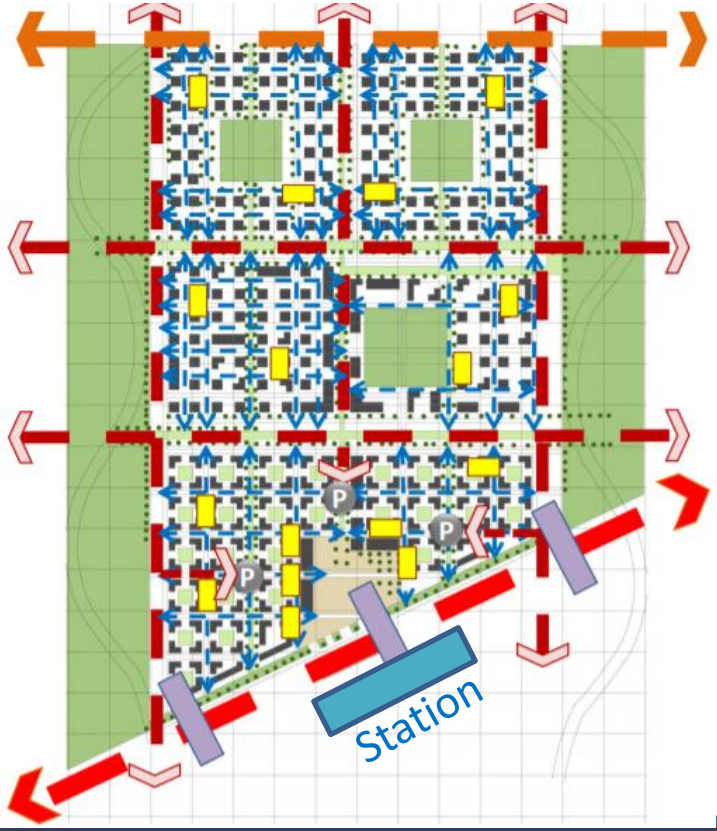
- Image showing the planned 3-D model for area around the metro station.
- Immediate area will have high rise development (5-8 floors building and plaza having 8-15 floors.)

Social Infrastructure



- Many social structures like schools, gyms, hotels, hospitals, clubs etc. are planned in nearby area.
- Partnership with private sector is important for such development.

Supporting Transport



National Highway 236

Circumventing Route

Motorised Network Including Bus Routes

Non Motorised Network

Pedestrian Crossing

Cycle Rickshaw Parking
Cycle Parking
Cycle Hire

P Parking Location

Pedestrian crossings, separate cycle parking and other facilities are provided to ensure maximum usage of non-motorized transport.

Impact of TOD-Population: Without TOD

	Developable Area (Ha) – Planned Area	Unplanned Area (Ha) Urban Villages	Total Area (Ha)	Planned Population @250persons per Ha on Developable Area*	Unplanned Population- Urban Villages*	Total Population
Scenario 2022	710	435	1145	177,500	82,564	260,064
Scenario 2032	1148	435	1583	287,000	92,336	***379,336***

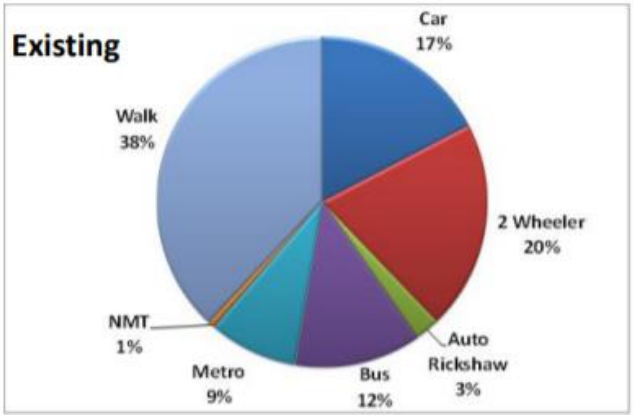
TOD minimizes the sprawl and tries to densify the development through compact & high-rise development.

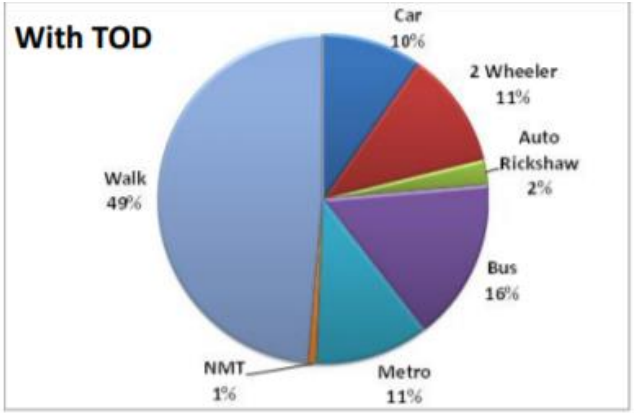
Impact of TOD-Population: TOD

	Urban Villages				Urban Villages	TOD 2022	TOD 2032	Total Population
	Chhatarpur	Sultanpur	Ghitorni	Ayanagar	Total			
					A	B	C	A+B+C
Census Year (2011)	30,574	13,704	11,036	16,792	72,107	-	-	72,107
Scenario 2022	33,902	15,141	12,198	21,324	82,564	641,826	-	724,390
Scenario 2032	37,458	16,709	13,463	24,705	92,336	704,725	513,461	1,310,522

TOD would have at least 65 to 70% of increase in the population in the Project area compared to the Planned Development of 2021

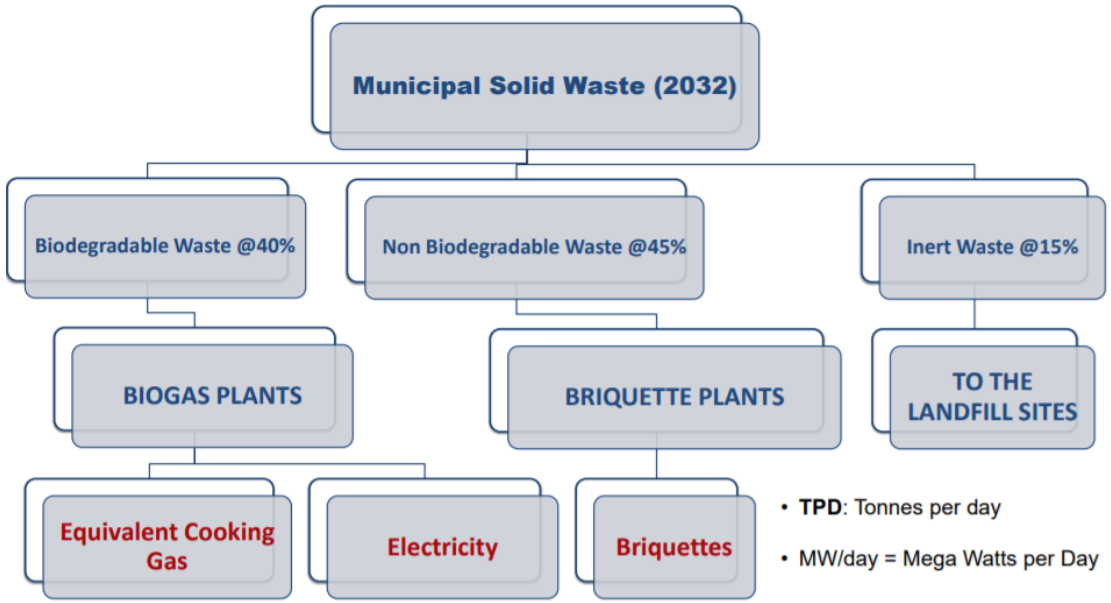
Impact of TOD: Modal Shift





- Public Transport + IPT share estimated to increase from 24% (current) to 29% (with TOD).
- Three-fold increase is observed in metro footfalls with TOD as compared to without TOD

Impact of TOD: Solid waste management



Under TOD, other aspects are also planned such as solid waste management facility for sustainable growth.

Impact of TOD: Water management

Water Supply

- Reduction in Water Demand through sewage water treatment & storm water harvesting
- Reclamation and Reuse of Sewage and Storm Water to reduce Water Demand

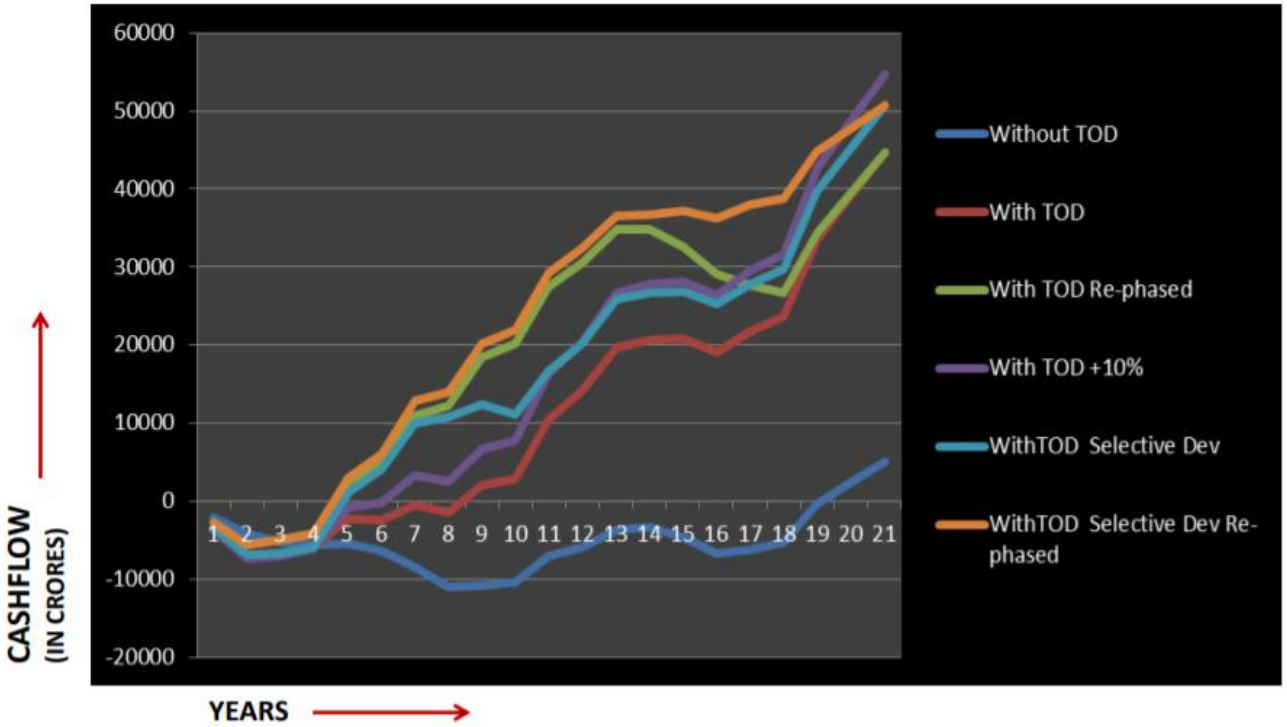
Sewage

- Grey Water is being reclaimed up to 85% for non potable purposes reducing the amount of sewage to STP's.

Storm Water Management

- 100% rain-water harvesting used to augment water supply.

TOD impact: Higher cash flow



- As per estimations, higher cash flow is estimated.
- Scenarios are compared such as with TOD, without TOD, rephased TOD etc.

Major challenges/risk

- At times, the powerful landowners may try to block project as per their personal interests.
- Cost of land acquisition may be driven by landowner expectations and lobbying.
- Major risk is the rate of sale of development - a slower rate than assumed this can very quickly turn a modest surplus into a significant deficit.

Conclusion

- TOD is a powerful tool for land use planning along the transit.
- High-rise and compact development through TOD can ensure accommodation of much more population along the transit thus providing socio-economic opportunities for more people.
- There are some possible risks may arise which need to be mitigated for smooth implementation of TOD.

References

- Delhi Metro Rail Project (DMRC), 2011 ‘Transit Oriented Development (TOD)-Study for existing metro corridor between Chhatarpur and Arjangarh of Delhi metro project of phase II’, 2011

- Delhi Development Authority (DDA), 2017 “Masterplan for Delhi-2021’

Thank You